EXPERIMENT – 39

Implementation of Letter Frequency Attack on Additive Cipher

CODE:

```
#include <stdio.h>
#include <ctype.h>

int main()
{
    char cipher[500], plain[500];
    int freq[26] = {0}, shift[26];
    int i, j, temp, top;

    printf("Enter ciphertext: ");
    fgets(cipher, sizeof(cipher), stdin);

    printf("Enter number of possible plaintexts: ");
    scanf("%d", &top);

    /* Count frequency */
    for (i = 0; cipher[i]; i++)
        if (isalpha(cipher[i]))
            freq[toupper(cipher[i]) - 'A']++;

    /* Sort shifts by frequency */
    for (i = 0; i < 26; i++)
        shift[i] = i;
```

```
for (i = 0; i < 25; i++)
    for (j = i + 1; j < 26; j++)
        if (freq[shift[i]] < freq[shift[j]])
        {
            temp = shift[i];
            shift[i] = shift[j];
            shift[j] = temp;
        }

printf("\nTop %d possible plaintexts:\n", top);

for (i = 0; i < top && i < 26; i++)
{
    int s = shift[i];

    printf("%d. ", i + 1);

    for (j = 0; cipher[j]; j++)
    {
        if (isalpha(cipher[j]))
        {
            int c = toupper(cipher[j]) - 'A';
            plain[j] = (c - s + 26) % 26 + 'A';
        }
        else
            plain[j] = cipher[j];

        printf("%c", plain[j]);
    }
```

```
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
C:\Users\deept\OneDrive\Doc
Enter ciphertext: KHOOR ZRUOG
Enter number of possible plaintexts: 10

Top 10 possible plaintexts:
1. WTAAD LDGAS

2. TQXXA IADXP

3. AXEEH PHKEW

4. EBIIL TLOIA

5. DAHHK SKNHZ

6. QNUUX FXAUM

7. LIPPS ASVPH

8. JGNNQ YQTNF

9. CZGGJ RJMGY

10. BYFFI QILFX


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Process exited after 15.72 seconds with return value 0
Press any key to continue . . .
```